

Yicheng Ke

+86 187-7315-2670 | keych@mail2.sysu.edu.cn | Changsha, Hunan, China
Undergraduate Student in Microelectronics Science and Engineering

Education

Sun Yat-sen University

B.Eng. Candidate, Microelectronics Science and Engineering

Sep 2024 - Jun 2028 (Expected)

Guangzhou, China

Average Score: 88/100 (GPA 3.8/4.0 top 15%); CET-4: 527; CET-6: 474.

Core Courses: Circuit Theory (90), College Physics (92), Probability and Statistics (93), Analog Circuits (91), Digital Circuits (91), Complex Variable Functions (96).

Research Experience

Temperature-Compensation Asymmetry at Thermal Extremes in 3D NAND Flash

Aug 2025 - Present

First-Author Researcher, 3D NAND Temperature-Compensation and TCAD Mechanism Study

- Identified cold-side undercompensation, high-state hot-side overcompensation, and read-window compression across WL regions, dies, pages, and retention-control tests.
- Linked piecewise T_{co} and state-dependent anomalies to trap emission, Poole-Frenkel charge loss, and mobility degradation using TC-disabled baselines and Sentaurus TCAD.

Mechanism-Informed V_{TH} /Sigma Prediction for 3D NAND Flash

Aug 2025 - May 2026

First-Author Researcher, 3D NAND Predictive Modeling and Controller-Oriented Reliability Study

- Reconstructed P1-P7 V_{TH} distributions from V_{ref} sweeps across -55 °C to 125 °C, extracting mean/sigma descriptors under cross-temperature program/read conditions.
- Built a mechanism-informed predictor with trap/barrier/mobility terms, improving extreme-temperature prediction and supporting controller-level V_{ref} optimization.

Domain-Density-Dependent Variability and Search Reliability in 2FeFET TCAM

Jan 2025 - Aug 2025

First-Author Researcher, FeFET Compact-Modeling and Array-Level Reliability Study

- Extended an NLS-FeFET Verilog-A compact model with domain-switching statistics, connecting device-level V_{TH} variation to 2FeFET-TCAM Monte Carlo search failures.
- Showed that higher ferroelectric domain density suppresses memory-window variation and search failures, guiding variability-aware embedded NVM design.

Publications

Journal Articles

- Yicheng Ke**, Minghui Dong, Linlin Cai, and **Wangyong Chen***. "A Mechanism-Informed Predictive Model for Mean and Sigma of 3D NAND Threshold-Voltage Distributions at Extreme Temperatures." *IEEE Electron Device Letters*, Early Access, 2026. doi: 10.1109/LED.2026.3704253.
- Yicheng Ke**, Minghui Dong, Linlin Cai, and **Wangyong Chen***. "Temperature Compensation Asymmetry at Thermal Extremes in 3D NAND Flash: Physical Mechanisms and State Dependence." *IEEE Transactions on Electron Devices*, Early Access, 2026, doi: 10.1109/TED.2026.3708111.
- Yicheng Ke**, et al.. "Domain Density Dependence in 2FeFET-Based TCAM Bit-Cell: Search Failure Analysis and Design Optimization." *IEEE Transactions on Device and Materials Reliability*. Early Access, 2026, doi: 10.1109/TDMR.2026.3707975.

Conference Papers

- Yicheng Ke**, Minghui Dong, Linlin Cai, and **Wangyong Chen***. "State-Dependent Overcompensation and Piecewise Temperature Response: New Observations from 3D NAND Cross-Temperature Characterization." *IEEE International Symposium on the Physical and Failure Analysis of Integrated Circuits (IPFA)*. (Accepted, 2026.)
- Yicheng Ke** and M. Li. "Comprehensive Variation Analysis of Domain Number and Write Pulse in NLS Theory-Based Ferroelectric Field-Effect Transistors." 2025 4th International Symposium on Semiconductor and Electronic Technology (ISSET), Xi'an, China, 2025, pp. 701–705.

* Corresponding author.

Subject Competition

2025 China Undergraduate Mathematical Contest in Modeling (CUMCM)

Provincial First Prize (Team Leader)

2025 Huashu Cup Mathematical Modeling Competition

Third Prize

Professional Skills

Simulation and EDA: Sentaurus TCAD, Cadence, Multisim.

Programming and Modeling: Python, C, MATLAB, Verilog-A, PyTorch.